野火_7寸_MIPI_LCD屏幕转接板_原理图_V0.1

目录

Page 1	目录
Page 2	历史版本
Page 3	接口
Page 4	电源驱动
Page 5	
Page 6	
Page 7	
Page 8	
Page 9	
Page 10	
Page 11	
Page 12	
Page 13	
Page 14	

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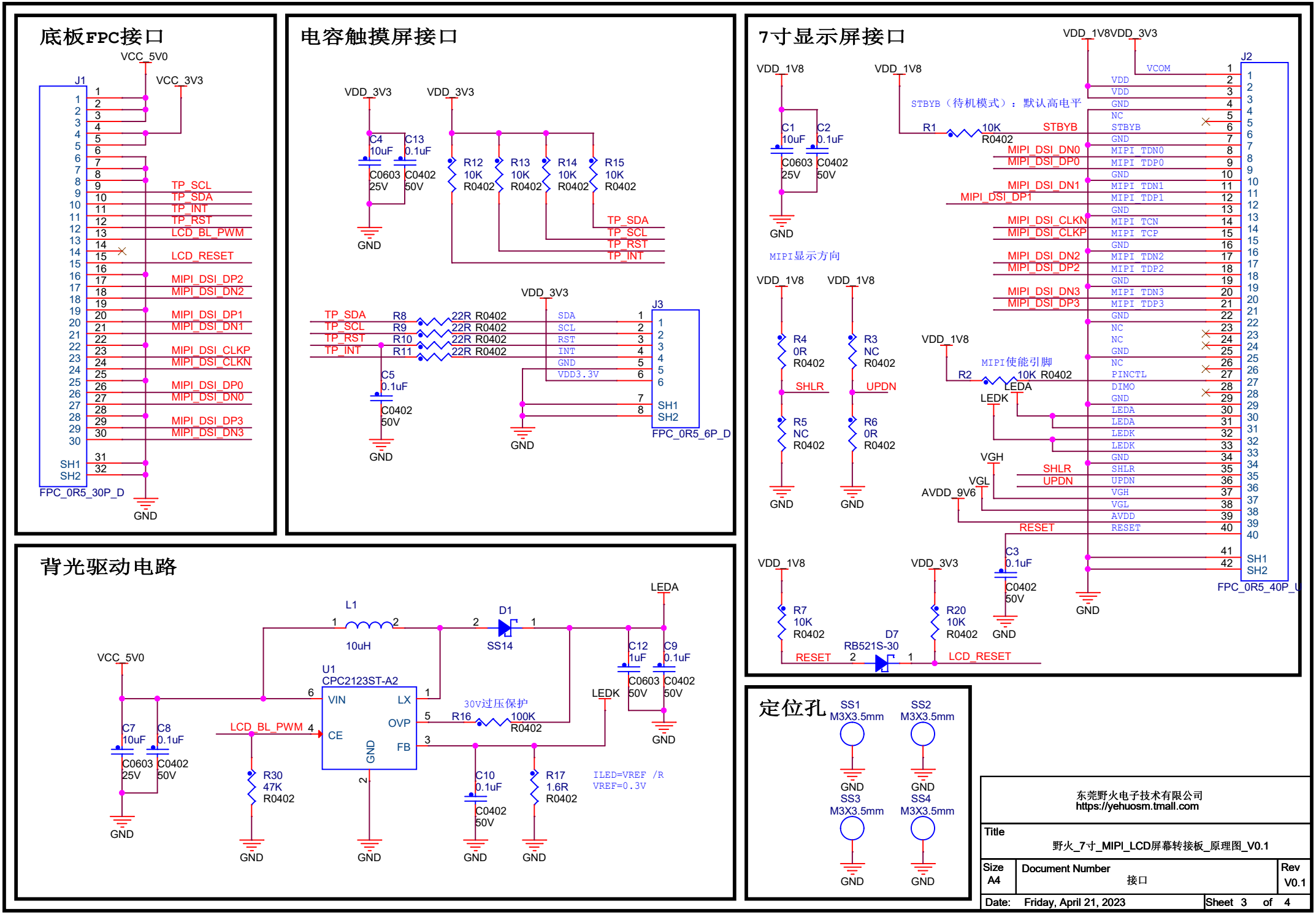
Title
野火_7寸_MIPI_LCD屏幕转接板_原理图_V0.1

Size A4 | Document Number 目录 | Rev V0.1

Date: Monday, April 10, 2023 | Sheet 1 of 4

历史版本

版本号	日期	设计	描述
V0.1	2023-02-17	ZTC	初始版本
V0.2	2023-04-20	ZTC	限流电阻改1.6R。R27、R28改1K，R26改47K，R29改10K


底板FPC接口
VCC_5V0
VCC_3V3
J1
FPC_0R5_30P_D
TP_SCL
TP_SDA
TP_INT
TP_RST
LCD_BL_PWM
LCD_RESET
MIPI_DSI_DP2
MIPI_DSI_DN2
MIPI_DSI_DP1
MIPI_DSI_DN1
MIPI_DSI_CLKP
MIPI_DSI_CLKN
MIPI_DSI_DP0
MIPI_DSI_DN0
MIPI_DSI_DP3
MIPI_DSI_DN3
SH1
SH2
GND
电容触摸屏接口
VDD_3V3
C4 10uF C0603 25V
C13 0.1uF C0402 50V
R12 10K R0402
R13 10K R0402
R14 10K R0402
R15 10K R0402
TP_SDA
TP_SCL
TP_RST
TP_INT
R8 22R R0402
R9 22R R0402
R10 22R R0402
R11 22R R0402
C5 0.1uF C0402 50V
J3
SDA
SCL
RST
INT
GND
VDD3.3V
SH1
SH2
FPC_0R5_6P_D
7寸显示屏接口
VDD_1V8
VDD_3V3
C1 10uF C0603 25V
C2 0.1uF C0402 50V
STBYB（待机模式）：默认高电平
R1 10K R0402
MIPI显示方向
R4 0R R0402
R3 NC R0402
R5 NC R0402
R6 0R R0402
SHLR
UPDN
MIPI使能引脚
R2 10K R0402
LEDA
LEDK
VGH
VGL
AVDD_9V6
RESET
C3 0.1uF C0402 50V
R7 10K R0402
R20 10K R0402
D7 RB521S-30
LCD_RESET
J2
VCOM
VDD
GND
NC
STBYB
MIPI_TDN0
MIPI_TDP0
MIPI_TDN1
MIPI_TDP1
MIPI_TCN
MIPI_TCP
MIPI_TDN2
MIPI_TDP2
MIPI_TDN3
MIPI_TDP3
PINCTL
DIMO
AVDD
SH1
SH2
FPC_0R5_40P_U
背光驱动电路
LEDA
L1 10uH
D1 SS14
C12 1uF C0603 50V
C9 0.1uF C0402 50V
VCC_5V0
U1 CPC2123ST-A2
VIN
LX
OVP
CE
FB
GND
30V过压保护
R16 100K R0402
LEDK
C7 10uF C0603 25V
C8 0.1uF C0402 50V
LCD_BL_PWM
R30 47K R0402
C10 0.1uF C0402 50V
R17 1.6R R0402
ILED=VREF /R
VREF=0.3V
定位孔
SS1 M3X3.5mm
SS2 M3X3.5mm
SS3 M3X3.5mm
SS4 M3X3.5mm
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Title
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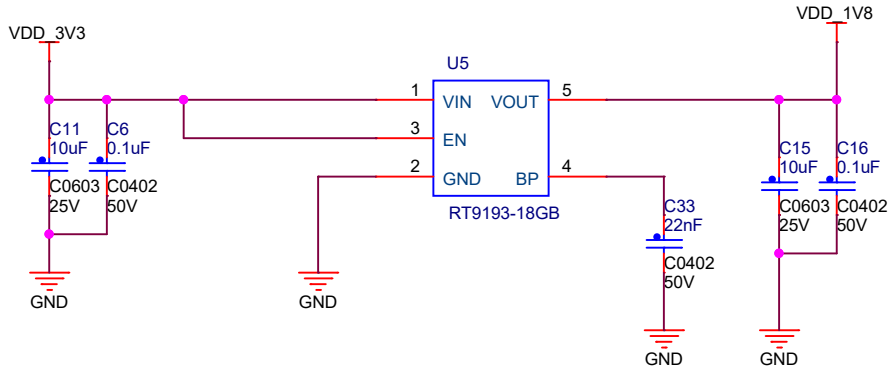
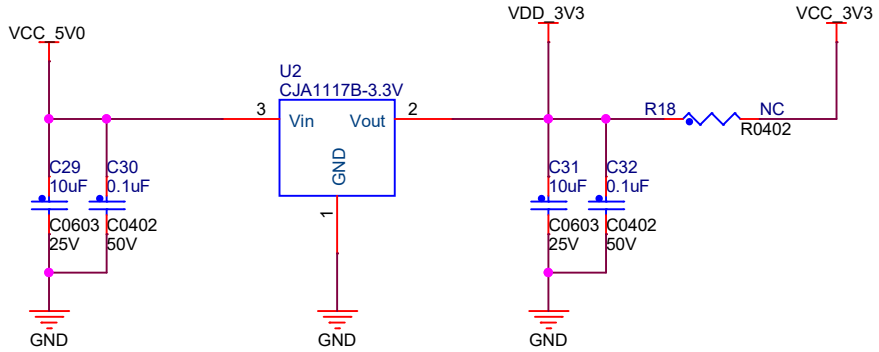
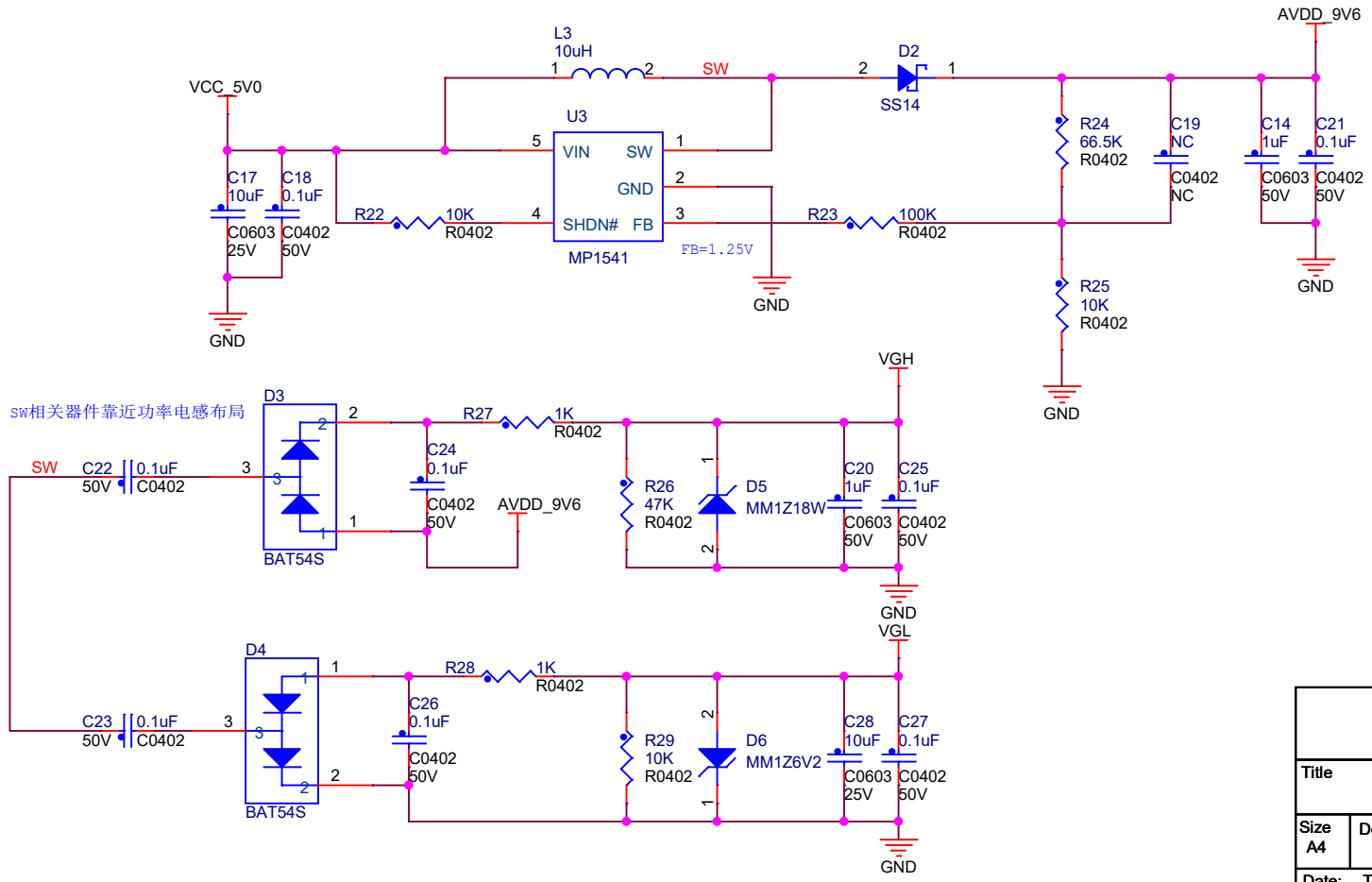
Size A4
Document Number
接口
Rev V0.1
Date: Friday, April 21, 2023
Sheet 3 of 4

降压1v8





电源电压



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Size A4	Document Number 电源驱动	Rev V0.1
Date: Thursday, April 20, 2023	Sheet 4 of 4	